Lakshman Raj Baskaran | Software Engineer

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Summary

Software Engineer with an M.Sc. in Computer Science from Trinity College Dublin, building **agentic AI systems**, **LLM evaluation pipelines** and the **Python and Java backends** behind them: **FastAPI**, **Spring Boot**, **PostgreSQL**, **Docker** and **Azure**.

Experience

AI Workflow Engineer Intern, FOREO — Sep 2026 - Nov 2026

- Building a tool that checks how well FOREO's published FAQs answer real customer questions, using **2,584** marketplace questions across **93** products: the FAQs answer only about **40%** of them, and **8 of the 10** biggest gaps it flagged were confirmed by manual review.
- Grouped similar questions into **16 topics** with multilingual **embeddings** and **HDBSCAN**, and used an **LLM** to judge whether an FAQ actually answers each one; its verdicts matched human labels **86.7%** of the time on held-out data.

Engineering Intern, Siemens Centre of Excellence — Aug 2023 - Sept 2023
Industrial Automation Systems, India

- Delivered a **warehouse management automation** project in **PLC ladder logic** inside a small engineering team, coordinating on shared control logic so that separately written pieces behaved correctly once combined.

Technical Skills

Languages: Python, Java, SQL, C++, R
AI and Agents: Anthropic Claude API, Deep Agents, Model Context Protocol (MCP), Spring AI, Ollama (Qwen), LangSmith, multi-agent systems, human-in-the-loop workflows, LLM evaluation, PyTorch, scikit-learn, HDBSCAN
Backend and Data: Spring Boot (REST, JPA, AOP), FastAPI, Flask, Pydantic, SQLAlchemy, PostgreSQL, SQLite, Redis, Azure Cosmos DB
Cloud and DevOps: Docker, GitHub Actions (CI/CD), Azure Container Apps, Kubernetes (minikube), Terraform (basic), Linux, Git
Testing: pytest, JUnit 5, AssertJ, Playwright

Projects

AI Agent System (Revenue Manager) | *Python, Deep Agents, Anthropic Claude API, PostgreSQL, Playwright, Chainlit*

- Built an **agentic AI revenue-manager** for hotels answering revenue queries in real time, using **7 tools, 2 subagents, and 11 domain skill files**, on the **Deep Agents framework** with **Claude Sonnet API** with **human-in-the-loop** approval.
- Cut latency **58% (57.5s → 24.1s)** by using **LangSmith tracing** to root-cause **context accumulation across shared threads**, and reduced **LLM calls** with a **deterministic routing table**.
- Built a full **data pipeline and UI**: **Playwright** web scraper, **ETL into PostgreSQL**, **Chainlit** front end, with **50 unit tests** across tools, pipeline, skills and agent wiring, including checks of tool output against direct SQL.

Spring Introspection MCP Server | *Java 21, Spring Boot 4, Spring AI 2, MCP*

- Built a **Spring Boot starter** for AI coding agents exposing **7 MCP tools**, resolving **property precedence chains** and **auto-configuration condition outcomes** that depend on **runtime env vars and classpath**, invisible in static source.
- Cut token usage **88% (80.7K → 9.5K tokens/query)** on **configuration and bean-provenance** questions by replacing multi-step **Actuator/source exploration** with **single tool calls**, benchmarked against a **naive source-reading agent**.

Disaster Management and Citizen Safety Platform | *Java Spring Boot, PostgreSQL, Next.js, Docker* — Sep 2024 - Mar 2025

- Built a **citizen safety platform** in a team of 8 which acted as a single source of truth during disasters, ingesting **incident reports**, dispatching the right **emergency services**, and **rerouting people and traffic** away from affected zones.
- Owned part of the **Spring Boot backend**; added a **deterministic fallback** that removed a **single point of failure** in resource allocation, and set up **CI/CD on GitHub Actions** building **Docker images**.

Education

Trinity College Dublin — 2024 - 2025
M.Sc. Computer Science, Future Networked Systems. **Thesis: Automatic Modulation Classification Using Graph Neural Networks**

SRM Institute of Science and Technology — 2020 - 2024
B.Tech. Electronics and Communication Engineering, 8.52/10. Research on photonic biosensors for skin cancer detection.

Indian Institute of Technology Madras — 2021 - 2026
Diploma in Programming: Data Structures and Algorithms, OOP in Java, Python, Modern Application Development.

Certifications and Publications

- Microsoft Certified: Azure AI Cloud Developer Associate, Aug 2026 · Model Context Protocol (MCP), Anthropic, Apr 2026.
- Co-authored a book chapter, Advanced Materials for Biomedical Devices (2024), on photonic biosensor design.